Yayılı Kuvvetler
Ağırlık Merkezi

Ağırlık ve ağırlık merkezi

- Dünya cismi oluşturan zerrelerin hepsine bir kuvvet uygular. O halde dünyanın bir rijit cisme etkisi, bütün cisim üzerine yayılmış çok sayıda kuvvetle gösterilmelidir. Tüm kuvvetlerinin bileşkesine ağırlık, uygulama noktasına da ağırlık merkezi denir.

Ne zaman Ağırlık Merkezi/Kütle Merkezini hesaplama ihtiyacı duyarız?

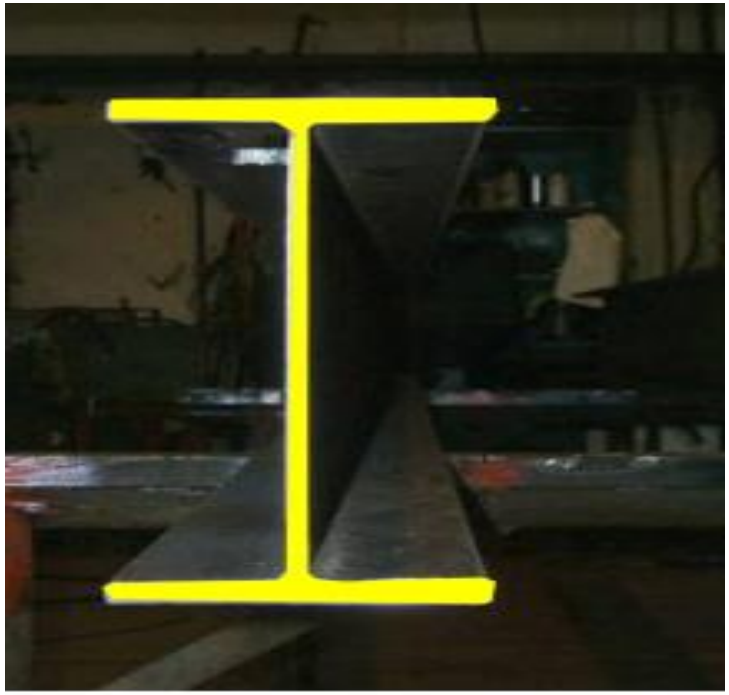
Statikte genellikle cismin toplam kütlesi ile ilgileniriz ve cismin toplam ağırlığını tekil bir kuvvet olarak göstermek isteriz



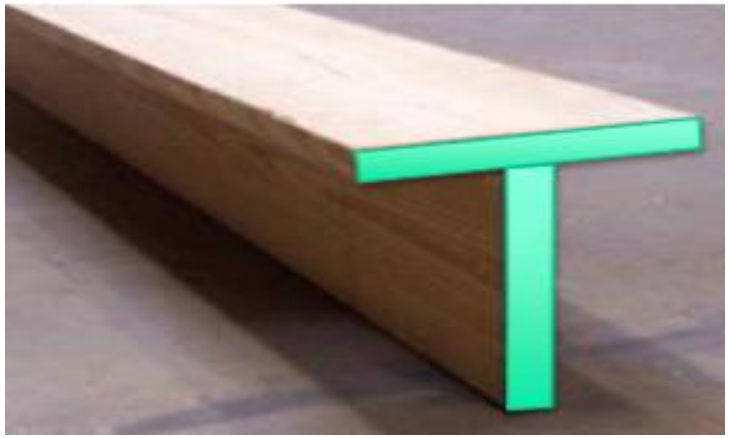
Su deposu taşıyıcı ayakların tasarımı



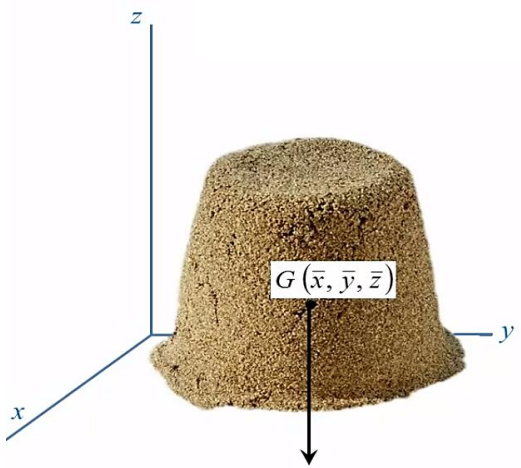
Aracın keskin virajlarda devrilmeye karşı daha emniyetli olması için kütle merkezi nerde olmalı?

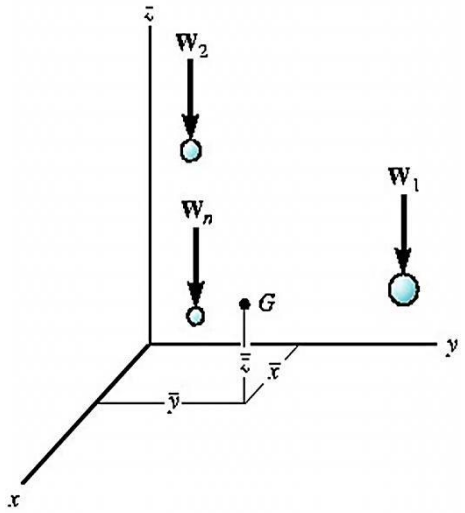


Çevremizde gördüğümüz birçok makine ve binalarda Taşıyıcı kirişler ve kolonlar kullanılır. Bu elemanların Kesitleri I, T ,U vb şeklinde olabilir.

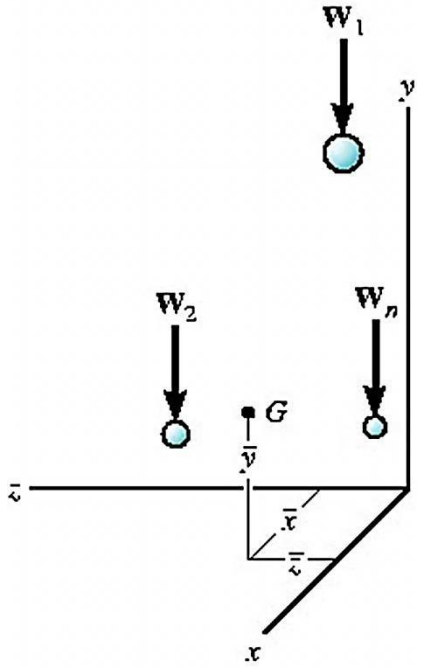


Bu kirişler taşıdıkları yük ve momentlerin tesiri ile şekil değiştirecekler ve gerilmeye maruz kalacaklardır. Gerilme ve şekil değişiminin hesaplanmasında kiriş kesitlerinin ağırlık merkezinin önemli tesiri bulunmaktadır.





(a)



(b)

$$W_R = \sum W$$

$$\bar{x} W_R = \tilde{x}_1 W_1 + \tilde{x}_2 W_2 + \ldots + \tilde{x}_n W_n$$

$$\bar{y} W_R = \tilde{y}_1 W_1 + \tilde{y}_2 W_2 + \ldots + \tilde{y}_n W_n$$

$$\bar{z} W_R = \tilde{z}_1 W_1 + \tilde{z}_2 W_2 + \ldots + \tilde{z}_n W_n$$

Ağırlık merkezi koordinatları

$$\bar{x} = \sum x \Delta W / \sum \Delta W$$

$$\bar{y} = \sum y \Delta W / \sum \Delta W$$

$$\bar{z} = \sum z \Delta W / \sum \Delta W$$

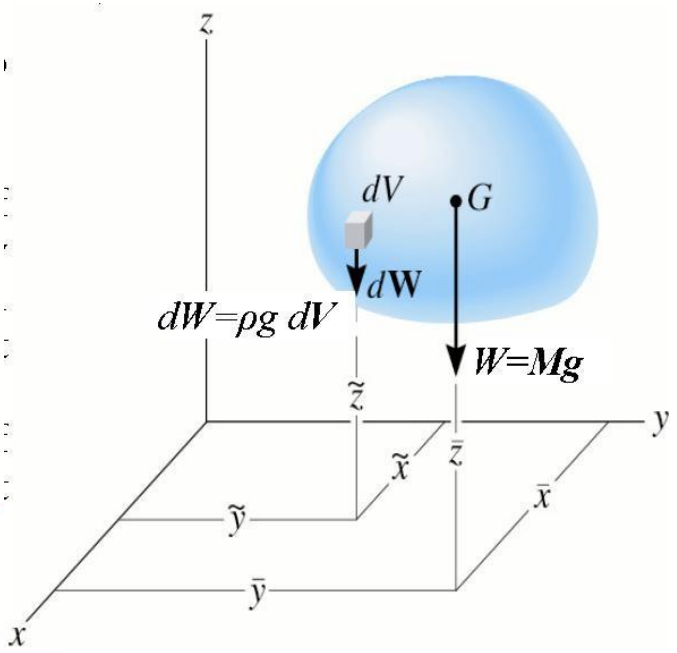
$$\bar{x} = \frac{\int x \cdot dW}{\int dW}$$

$$\bar{y} = \frac{\int y \cdot dW}{\int dW}$$

$$\bar{z} = \frac{\int z \cdot dW}{\int dW}$$

Yerçekimi kuvvetini her yerde aynı kabul ederek

Kütle merkezinin koordinatları elde edilir.



$$W = mg$$

$$\bar{x} = \frac{\int x \cdot dW}{\int dW} = \frac{\int x \cdot dm \cdot g}{\int dm \cdot g} = \frac{\int x \cdot dm}{\int dm}$$

$$\bar{x} = \frac{\int x \cdot dm}{\int dm} \qquad \bar{y} = \frac{\int y \cdot dm}{\int dm} \qquad \bar{z} = \frac{\int z \cdot dm}{\int dm}$$

Cisim homojen ve her yerde yoğunluk aynı ise;

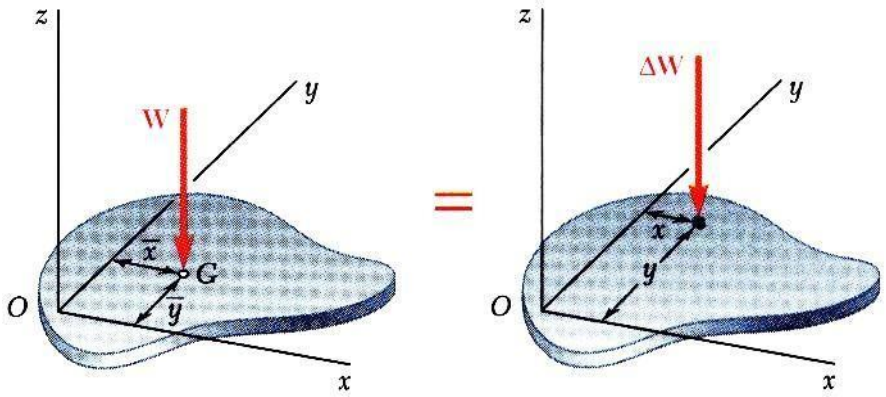
Geometri merkezinin koordinatları elde edilir. Bir hacmin ağırlık merkezi

$$m = \rho \cdot V$$ sabit

$$\bar{x} = \frac{\int x \cdot dm}{\int dm} = \frac{\int x \cdot \rho \cdot dV}{\int \rho \cdot dV} = \frac{\int x \cdot dV}{\int dV}$$

$$\bar{x} = \frac{\int x \cdot dV}{\int dV} \qquad \bar{y} = \frac{\int y \cdot dV}{\int dV} \qquad \bar{z} = \frac{\int z \cdot dV}{\int dV}$$

Cismin kalınlığı homojen ve her yerde aynı ise; geometri merkezinin koordinatları elde edilir. Bir alanın ağırlık merkezi



Alansal cisim

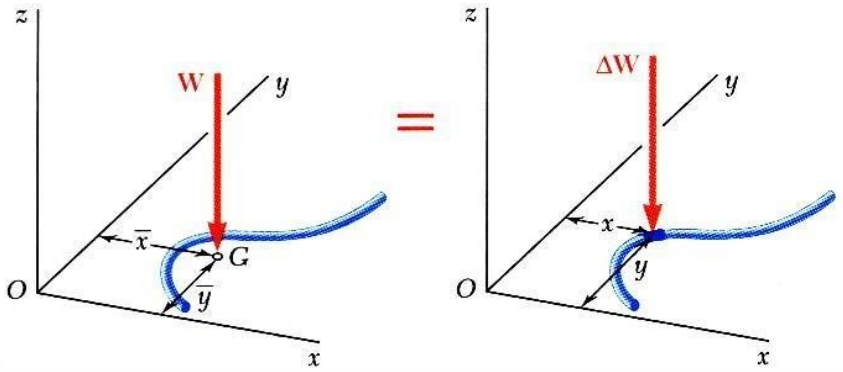
$$V = t \cdot A$$ sabit

$$\bar{x} = \frac{\int x \cdot dV}{\int dV} = \frac{\int x \cdot t \cdot dA}{\int t \cdot dA} = \frac{\int x \cdot dA}{\int dA}$$

$$\bar{x} = \frac{\int x \cdot dA}{\int dA} \qquad \bar{y} = \frac{\int y \cdot dA}{\int dA}$$

Cismin kesiti uzunluğu yanında çok küçük homojen ve her yerde aynı ise;

Geometri merkezinin koordinatları elde edilir. Bir çizgisel cismin ağırlık merkezi



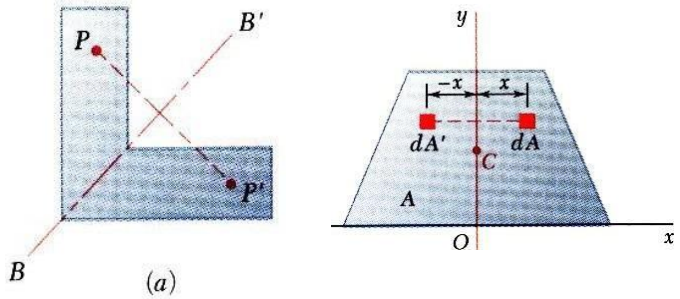
Çizgisel cisim

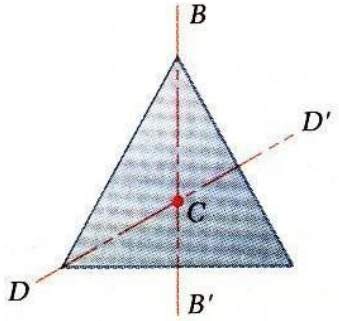
$$A = w \cdot l$$ Sabit kalınlık

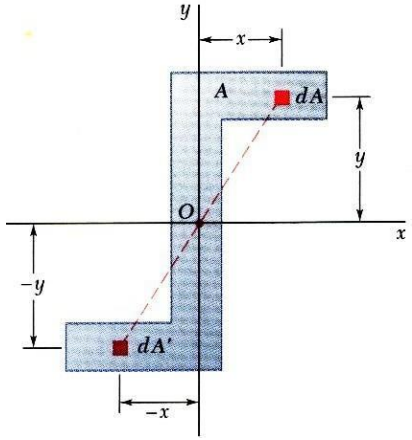
$$\bar{x} = \frac{\int x \cdot dA}{\int dA} = \frac{\int x \cdot w \cdot dl}{\int w \cdot dl}$$

$$\bar{x} = \frac{\int x \cdot dl}{\int dl} \qquad \bar{y} = \frac{\int y \cdot dl}{\int dl}$$

Alanların ve çizgilerin birinci momenti

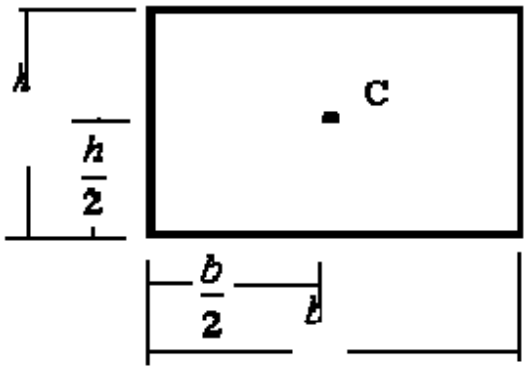


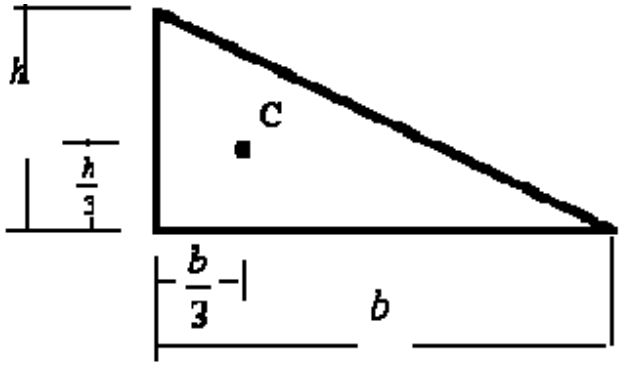


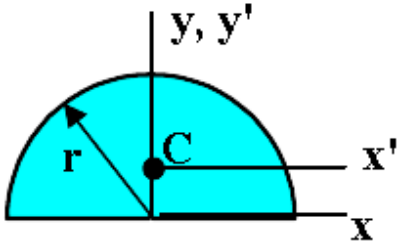


- Eğer bir alan BB' eksenine göre simetrik ise her P noktasına karşılık bir P' noktası vardır, PP' doğrusu BB' eksenine diktir ve BB'doğrusu tarafından ikiye bölünmüştür.
- Simetri eksenine göre alanın 1. momenti sıfırdır.
- Eğer bir alanın simetri ekseni varsa ağırlık merkezi bu eksen üzerindedir.
- İki simetri ekseni varsa, merkez onların kesişimindedir.
- Alanın merkezi simetri ekseni üzerindedir.

Çok kullanılan alan biçimlerinin ağırlık merkezleri



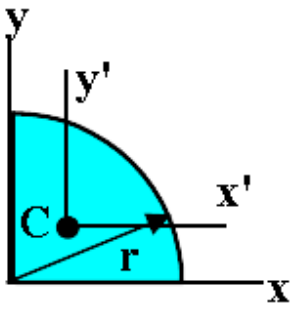




$$\bar{x} = 0$$

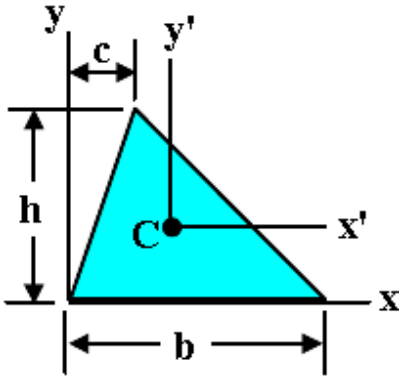
$$\bar{y} = \frac{4r}{3\pi}$$

$$A = \tfrac{1}{2}\pi r^2$$



$$\bar{x} = \bar{y} = \frac{4r}{3\pi}$$

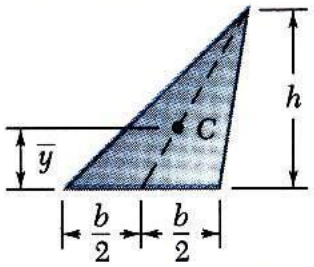
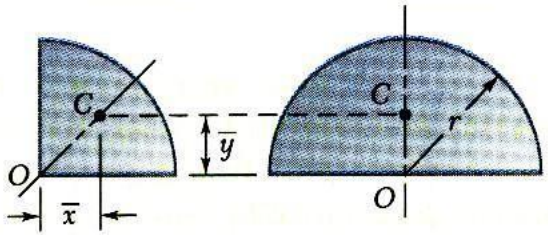
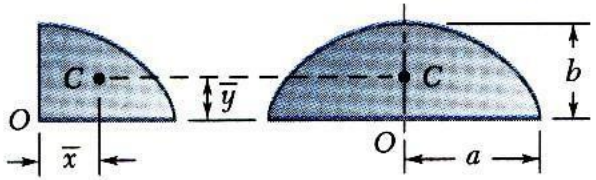
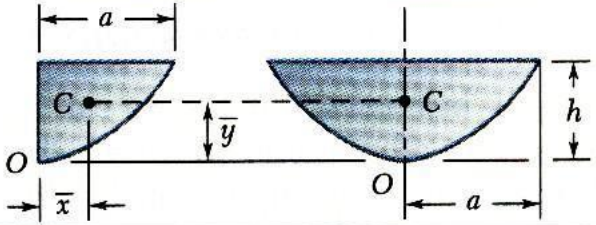
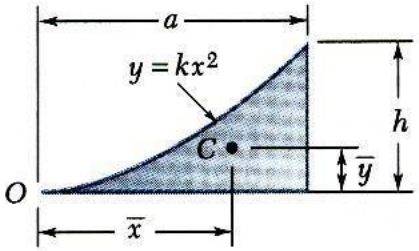
$$A = \tfrac{1}{4}\pi r^2$$

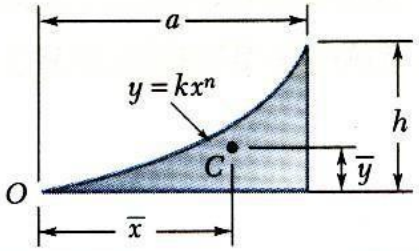
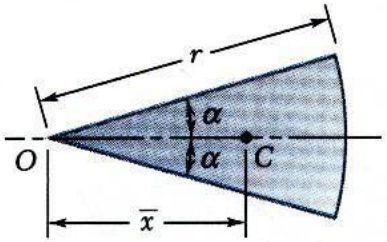


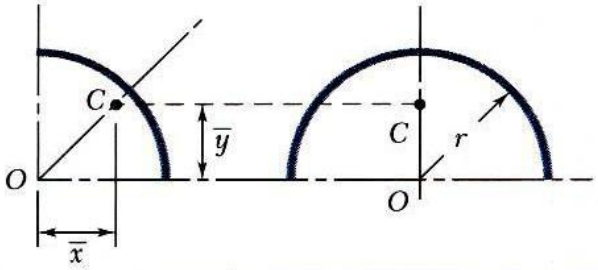
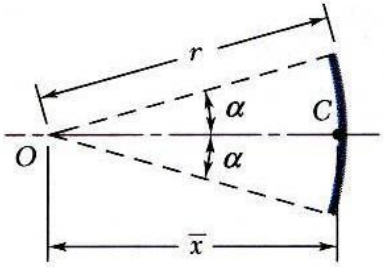
$$\bar{x} = \tfrac{1}{3}(b + c)$$

$$\bar{y} = \tfrac{1}{3}h$$

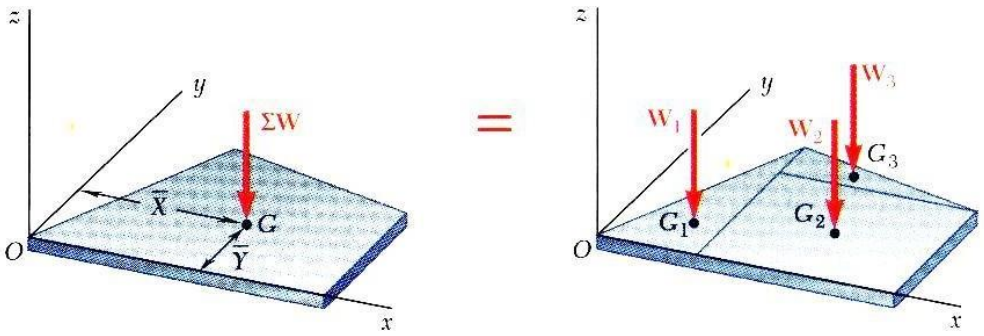
$$A = \tfrac{1}{2}bh$$

Shape		$\bar{x}$	$\bar{y}$	Area
Üçgen			$\frac{h}{3}$	$\frac{bh}{2}$
Dörttebir daire		$\frac{4r}{3\pi}$	$\frac{4r}{3\pi}$	$\frac{\pi r^2}{4}$
Yarım daire		0	$\frac{4r}{3\pi}$	$\frac{\pi r^2}{2}$
Dörttebir elips		$\frac{4a}{3\pi}$	$\frac{4b}{3\pi}$	$\frac{\pi ab}{4}$
Yarım elips		0	$\frac{4b}{3\pi}$	$\frac{\pi ab}{2}$
Yarım parabol		$\frac{3a}{8}$	$\frac{3h}{5}$	$\frac{2ah}{3}$
Parabol		0	$\frac{3h}{5}$	$\frac{4ah}{3}$
Parabol parçası		$\frac{3a}{4}$	$\frac{3h}{10}$	$\frac{ah}{3}$

Parabol parçası		$\frac{n+1}{n+2}a$	$\frac{n+1}{4n+2}h$	$\frac{ah}{n+1}$
Daire dilimi		$\frac{2r\sin\alpha}{3\alpha}$	0	αr^2

Dörttebir Daire yayı		$\frac{2r}{\pi}$	$\frac{2r}{\pi}$	$\frac{\pi r}{2}$
Yarım Daire yayı		0	$\frac{2r}{\pi}$	πr
Daire yayı		$\frac{r\sin\alpha}{\alpha}$	0	$2\alpha r$

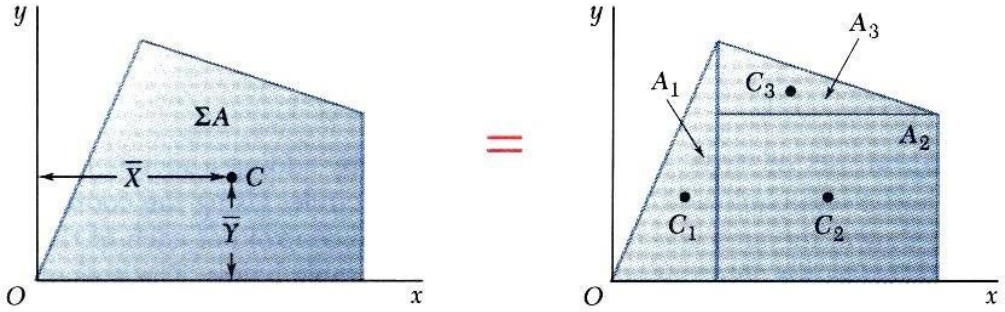
Bileşik Plak ve Teller



- Bileşik plaklar

$$\bar{X}\sum W = \sum \tilde{x} W$$

$$\bar{Y}\sum W = \sum \tilde{y} W$$

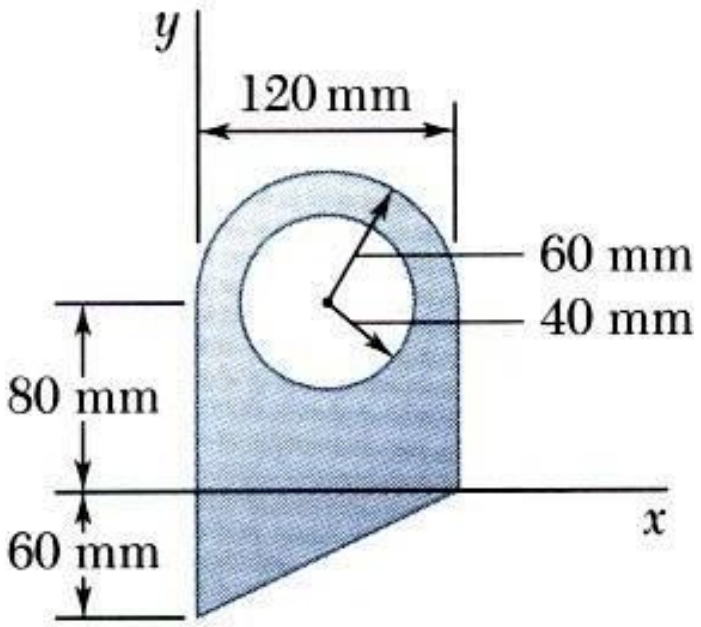


- Bileşik alanlar

$$\bar{X}\sum A = \sum \tilde{x} A$$

$$\bar{Y}\sum A = \sum \tilde{y} A$$

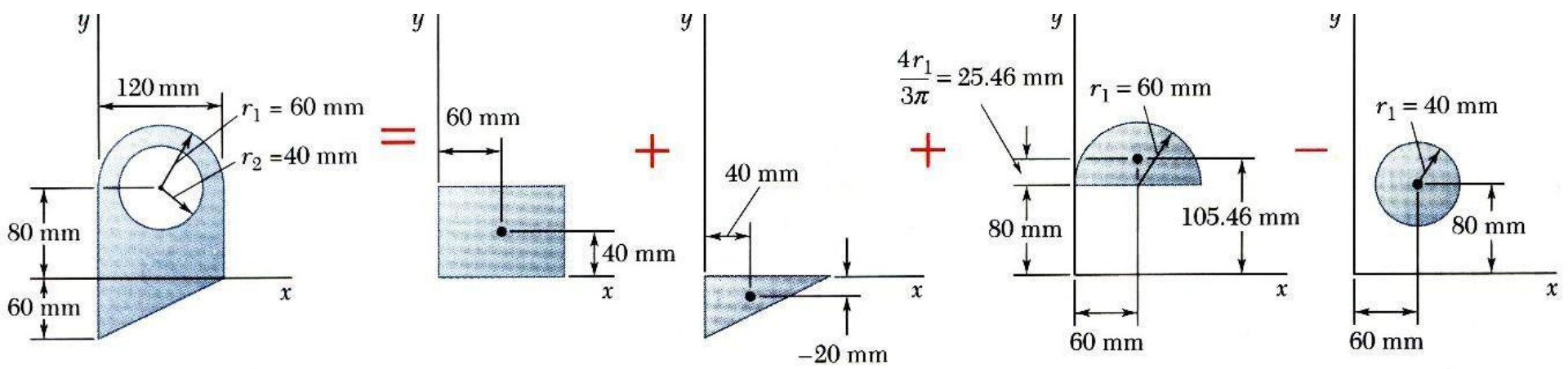
Problem



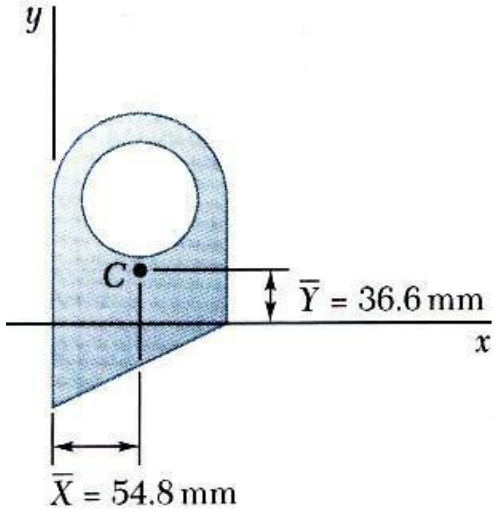
Yukardaki taralı alanın, x and y eksenlerine göre 1. momentlerini ve ağırlık merkezinin yerini bulunuz.

ÇÖZÜM:

- Alan, bir üçgen, bir dikdörtgen ve bir yarım dairenin toplamı ve bir dairenin çıkartılması şeklinde parçalara bölünmüştür.
- Her bir alanın eksenlere göre birinci momentleri hesaplanır.
- Üçgenin, dikdörtgenin ve yarım dairenin birinci momentleri toplanır ve dairenin birinci momenti bu değerden çıkartılır.
- Alanın merkezi, birinci momentlerin toplam alana bölünmesiyle elde edilir.



Component	A, mm^2	$\bar{x}$, mm	$\bar{y}$, mm	$\bar{x}A$, mm^3	$\bar{y}A$, mm^3
Rectangle	$(120)(80) = 9.6 \times 10^3$	60	40	$+576 \times 10^3$	$+384 \times 10^3$
Triangle	$\frac{1}{2}(120)(60) = 3.6 \times 10^3$	40	−20	$+144 \times 10^3$	-72×10^3
Semicircle	$\frac{1}{2}\pi(60)^2 = 5.655 \times 10^3$	60	105.46	$+339.3 \times 10^3$	$+596.4 \times 10^3$
Circle	$-\pi(40)^2 = -5.027 \times 10^3$	60	80	-301.6×10^3	-402.2×10^3
	$\Sigma A = 13.828 \times 10^3$			$\Sigma \bar{x}A = +757.7 \times 10^3$	$\Sigma \bar{y}A = +506.2 \times 10^3$



$$\bar{X} = \frac{\sum \bar{x}A}{\sum A} = \frac{+757.7\times10^3\,\text{mm}^3}{13.828\times10^3\,\text{mm}^2}$$

$$\bar{X} = 54.8\,\text{mm}$$

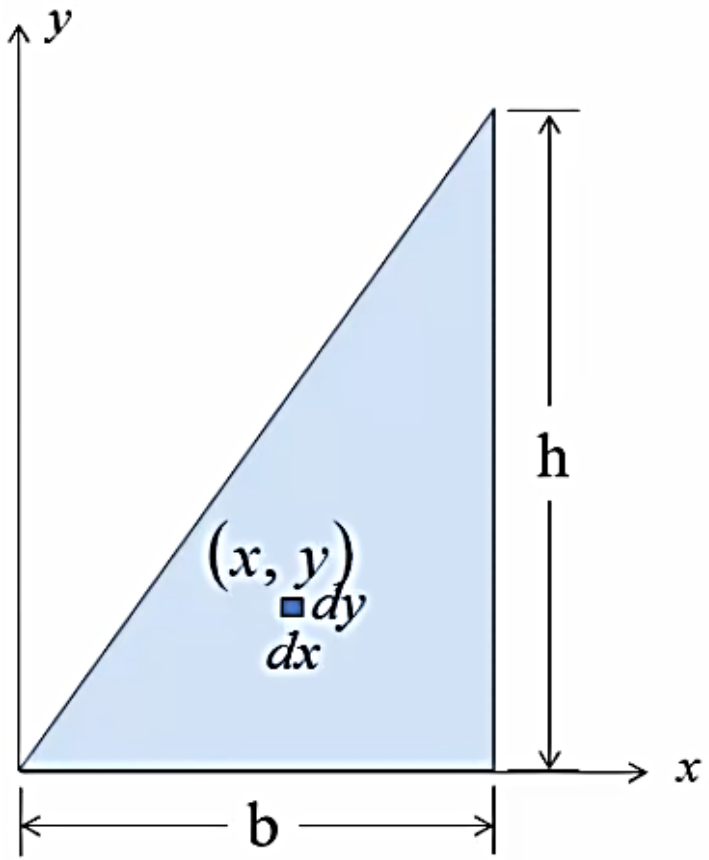
$$\bar{Y} = \frac{\sum \bar{y}A}{\sum A} = \frac{+506.2\times10^3\,\text{mm}^3}{13.828\times10^3\,\text{mm}^2}$$

$$\bar{Y} = 36.6\,\text{mm}$$

$$Q_x = +506.2\times10^3\,\text{mm}^3$$

$$Q_y = +757.7\times10^3\,\text{mm}^3$$

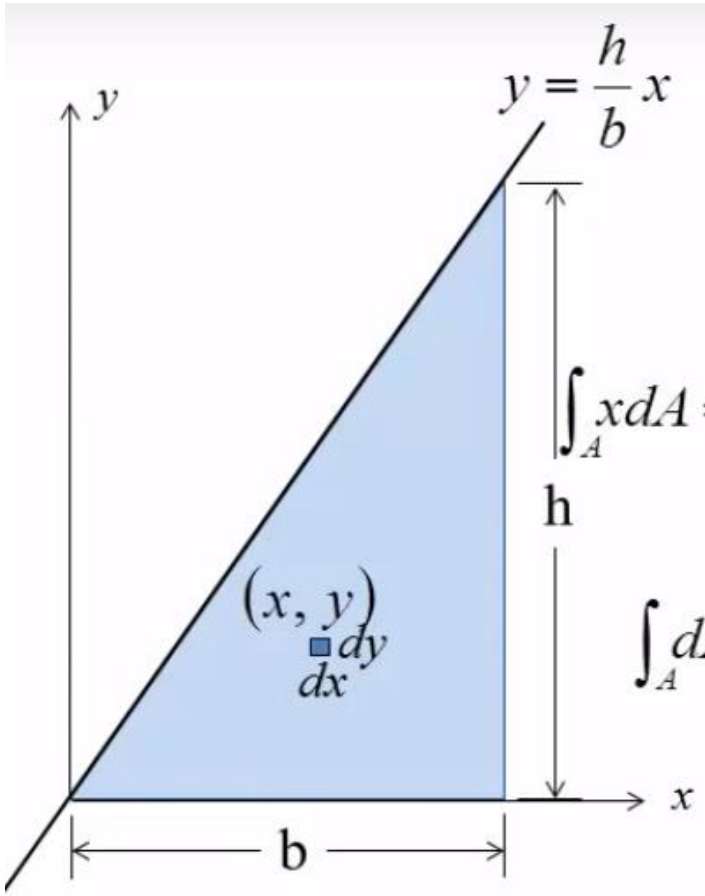
Ağırlık merkezinin integralle bulunması



$$\bar{x} = \frac{\int x \cdot dA}{\int dA}$$

$$\bar{y} = \frac{\int y \cdot dA}{\int dA}$$

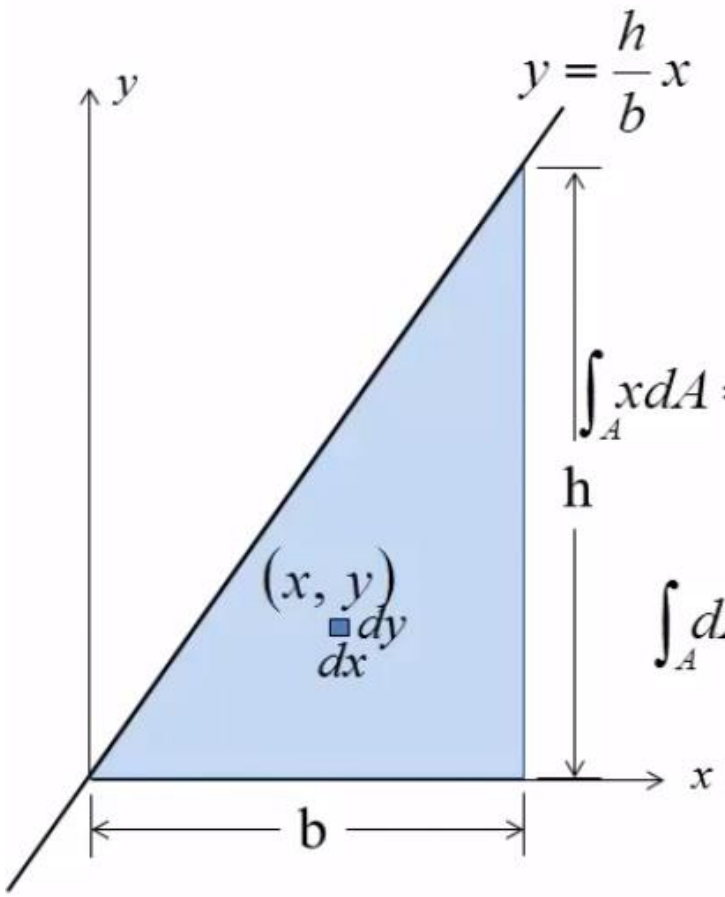
$$dA = dx \cdot dy$$



$$\bar{x} = \frac{\int x \cdot dA}{\int dA}$$

$$\int_A x dA = \int_0^b \int_0^{\frac{h}{b}x} x dy dx = \int_0^b \frac{h}{b} x^2 dx = \frac{h}{b} \cdot \frac{1}{3} x^3 \Big|_0^b = \frac{1}{3} hb^2$$

$$\int_A dA = \int_0^b \int_0^{\frac{h}{b}x} dy dx = \int_0^b \frac{h}{b} x dx = \frac{h}{b} \cdot \frac{1}{2} x^2 \Big|_0^b = \frac{1}{2} hb$$



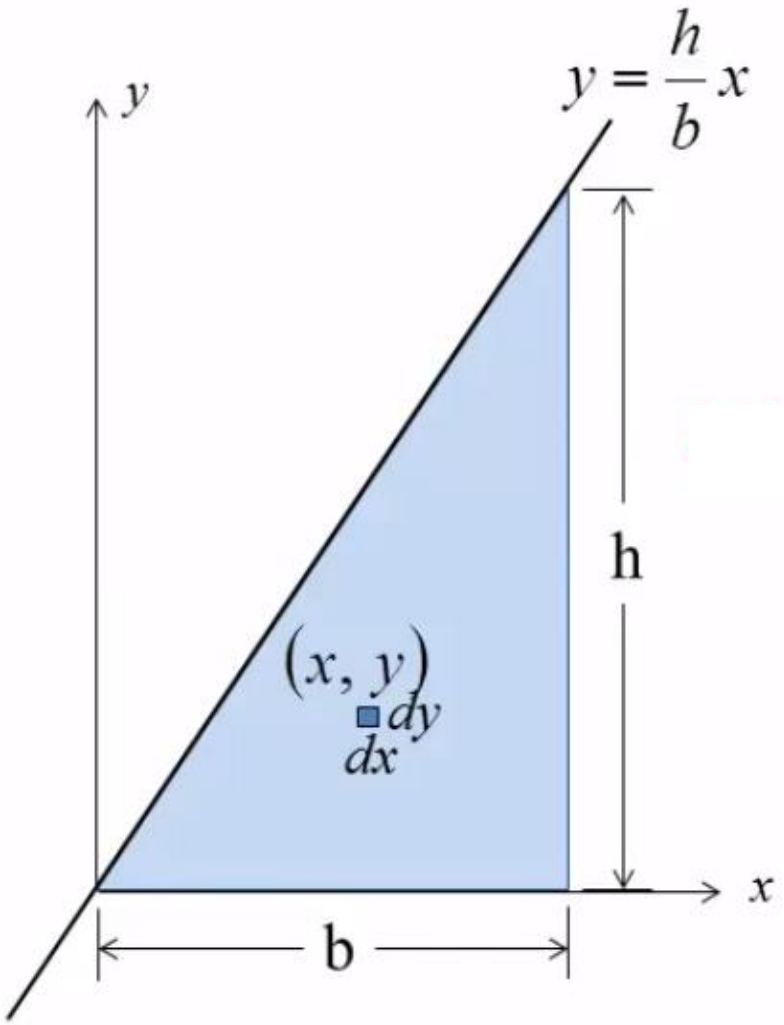
$$\bar{x} = \frac{\int x \cdot dA}{\int dA}$$

$$\int_A x dA = \int_0^b \int_0^{\frac{h}{b}x} x dy dx = \int_0^b \frac{h}{b} x^2 dx = \frac{h}{b} \cdot \frac{1}{3} x^3 \Big|_0^b = \frac{1}{3} hb^2$$

$$\int_A dA = \int_0^b \int_0^{\frac{h}{b}x} dy dx = \int_0^b \frac{h}{b} x dx = \frac{h}{b} \cdot \frac{1}{2} x^2 \Big|_0^b = \frac{1}{2} hb$$

Üçgenin alanı

$$\bar{x} = \frac{\int x \cdot dA}{\int dA} = \frac{2}{3} b$$



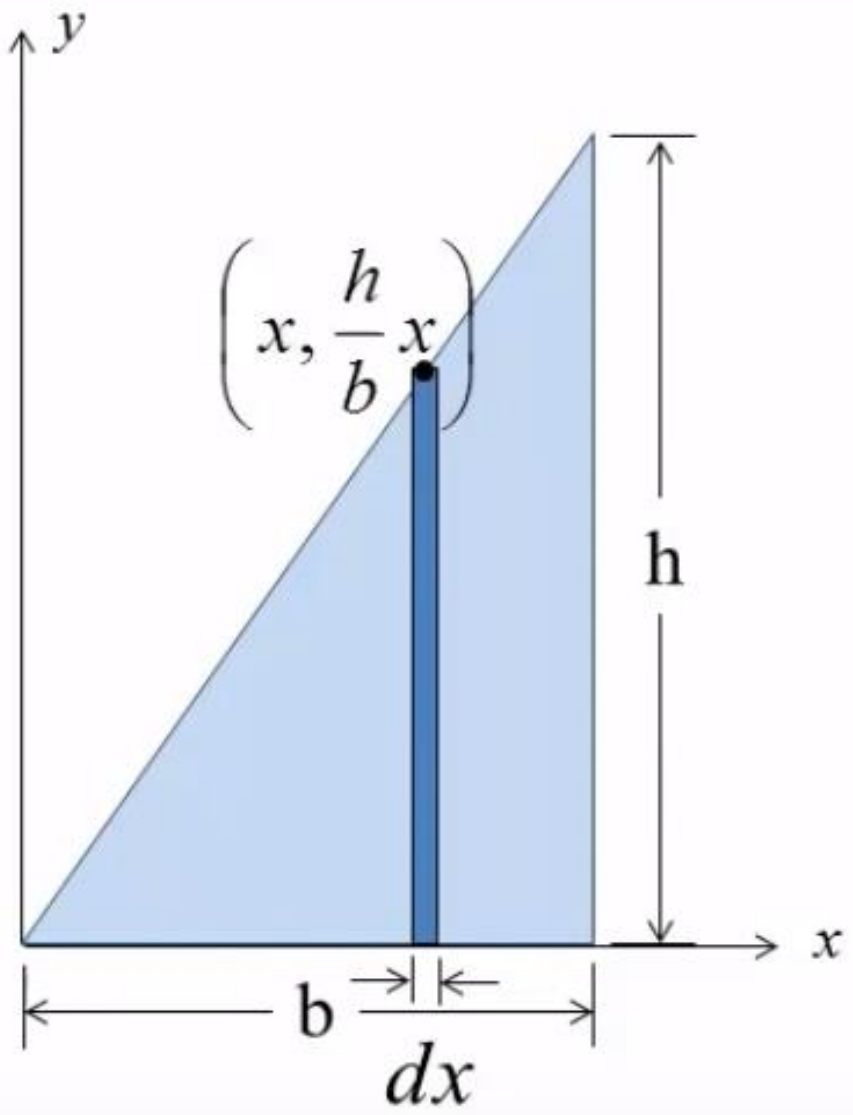
$$\bar{y} = \frac{\int y \cdot dA}{\int dA}$$

$$\int_A dA = \frac{1}{2}hb$$

$$\int_A y dA = \int_0^b \int_0^{\frac{h}{b}x} y dy dx = \int_0^b \frac{h^2}{2b^2} x^2 dx$$

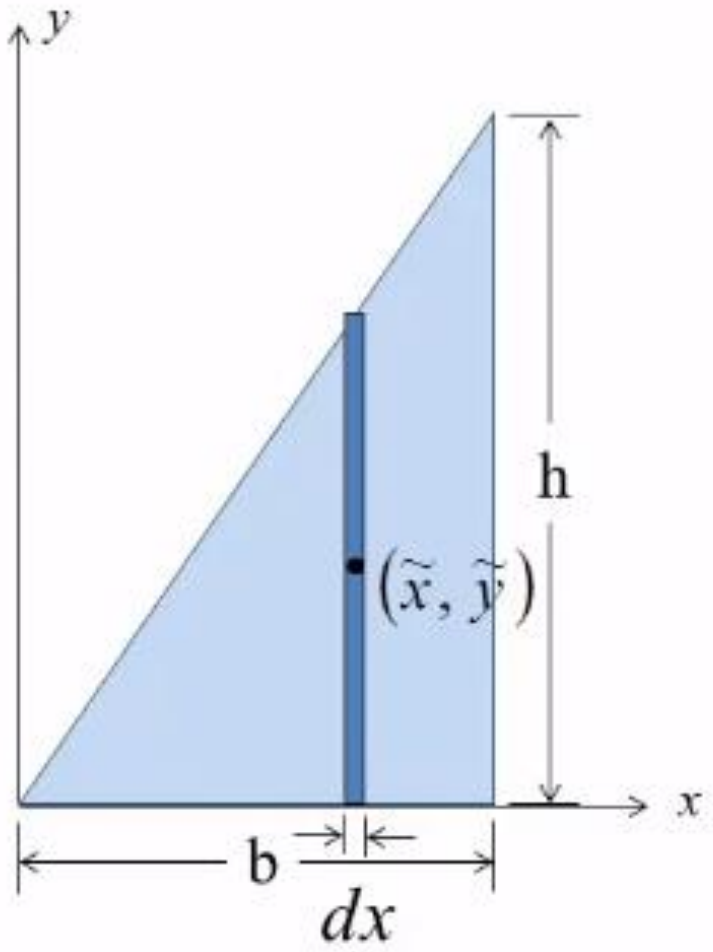
$$= \frac{h^2}{b^2} \cdot \frac{1}{6} x^3 \Big|_0^b = \frac{1}{6} h^2 b$$

$$\bar{y} = \frac{\int y \cdot dA}{\int dA} = \frac{1}{3}h$$



$$\bar{x} = \frac{\int \tilde{x} \cdot dA}{\int dA}$$

$$\bar{y} = \frac{\int \tilde{y} \cdot dA}{\int dA}$$

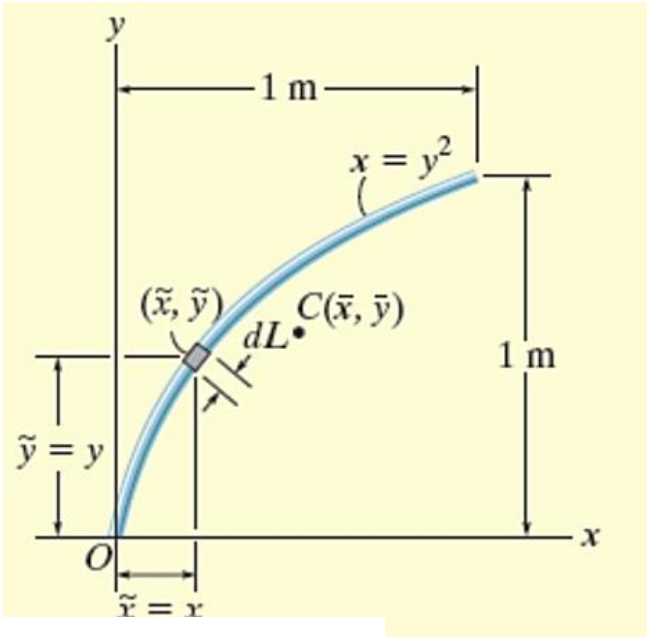


$$\tilde{x} = x \qquad dA = \frac{h}{b} x dx$$

$$\bar{x} = \frac{\int \tilde{x} \cdot dA}{\int dA} = \frac{\int x \cdot \frac{h}{b} x dx}{\int \frac{h}{b} x dx}$$

$$= \frac{\int x \cdot x dx}{\int x dx} = \frac{\int x^2 dx}{\int x dx} = \frac{\left. \frac{1}{3} x^3 \right|_0^b}{\left. \frac{1}{2} x^2 \right|_0^b} = \frac{2}{3} b$$

Parabolik şekildeki demir çubuğun ağırlık merkezinin yerini bulunuz.



$$dL = \sqrt{(dx)^2 + (dy)^2} = \sqrt{\left(\frac{dx}{dy}\right)^2 + 1}\,dy$$

$x = y^2$ $\qquad$ $dx/dy = 2y$

$$dL = \sqrt{(2y)^2 + 1}\,dy$$

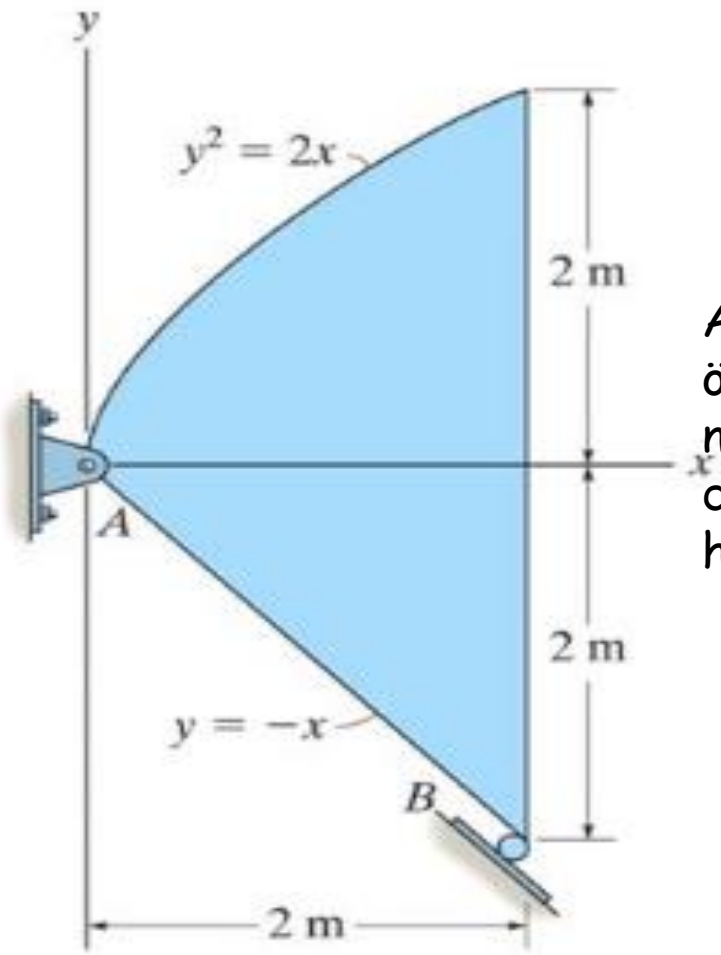
$\tilde{x} = x, \tilde{y} = y$

$$\bar{x} = \frac{\int_L \tilde{x}\,dL}{\int_L dL} = \frac{\int_0^1 x\sqrt{4y^2+1}\,dy}{\int_0^1 \sqrt{4y^2+1}\,dy} = \frac{\int_0^1 y^2\sqrt{4y^2+1}\,dy}{\int_0^1 \sqrt{4y^2+1}\,dy}$$

$$= \frac{0.6063}{1.479} = 0.410m$$

$$\bar{y} = \frac{\int_L \tilde{y}\,dL}{\int_L dL} = \frac{\int_0^1 y\sqrt{4y^2+1}\,dy}{\int_0^1 \sqrt{4y^2+1}\,dy}$$

$$= \frac{0.8484}{1.479} = 0.574m$$



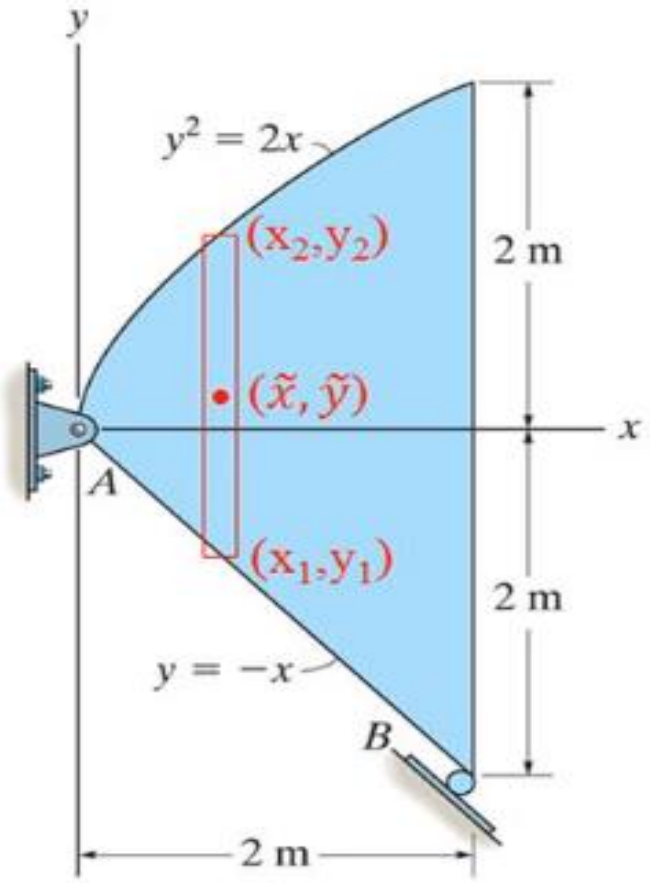
A noktasından mafsallanmış 0.3 m kalınlığında özkütlesi 7850 kg/m^3 şekildeki Plaka B noktasında sürtünmesiz mesnetle desteklenmektedir. Mesnet tepkilerini hesaplayınız.

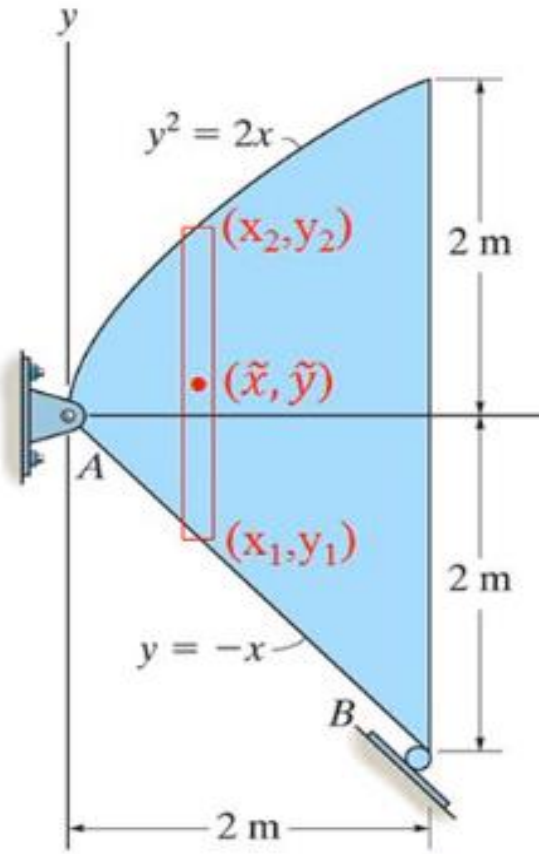
Çözüm

1. Plaka üzerindeki parçacığın alanı dA yı bulalım

2. $$dA = (y_2 - y_1)\, dx$$
$$= (\sqrt{2x} + x)\, dx$$

3. $$\tilde{x} = x$$
$$\tilde{y} = (y_1 + y_2)/2$$
$$= (\sqrt{2x} - x)/2$$





$$\bar{x} = \frac{\int_A \tilde{x}\,dA}{\int_A dA} = \frac{\int_0^2 x(\sqrt{2x}+x)dx}{\int_0^2 (\sqrt{2x}+x)dx} = \frac{\left[\left(\frac{2\sqrt{2}}{5}\right)x^{\frac{5}{2}} + \frac{1}{3}x^3\right]_0^2}{\left[\left(\frac{2\sqrt{2}}{3}\right)x^{\frac{3}{2}} + \frac{1}{2}x^2\right]_0^2}$$

$$= \frac{5.867}{4.667} = 1.257 \text{ m}$$

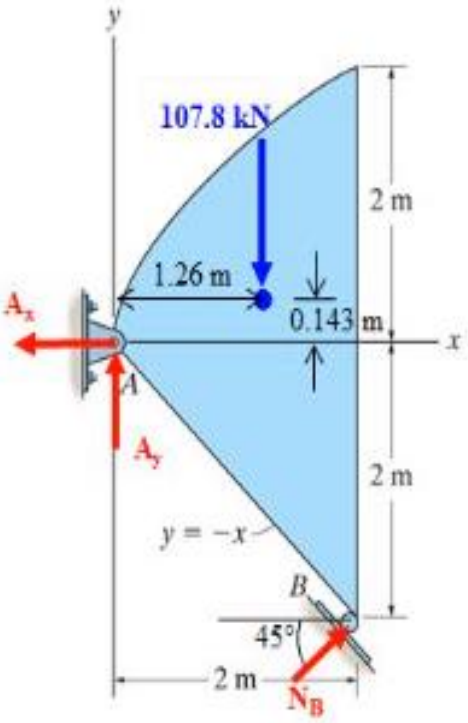
$$\bar{y} = \frac{\int_A \tilde{y}\,dA}{\int_A dA} = \frac{\int_0^2 \{(\sqrt{2x}-x)/2\}(\sqrt{2x}+x)dx}{\int_0^2 (\sqrt{2x}+x)dx} = \frac{\left[\frac{x^2}{2} - \frac{1}{6}x^3\right]_0^2}{\left[\left(\frac{2\sqrt{2}}{3}\right)x^{\frac{3}{2}} + \frac{1}{2}x^2\right]_0^2}$$

$$= \frac{0.6667}{4.667} = 0.143 \text{ m}$$

$\bar{x} = 1.26$ m and $\bar{y} = 0.143$ m

$$A = 4.667\ m^2$$

$$W = (7850)\,(9.81)\,(4.667)\,0.3 = 107.8\ kN$$



$$\circlearrowleft + \sum M_A = N_B\,(2\sqrt{2}) - 107.8\,(1.26) = 0$$

$$N_B = 47.92 = 47.9\ kN$$

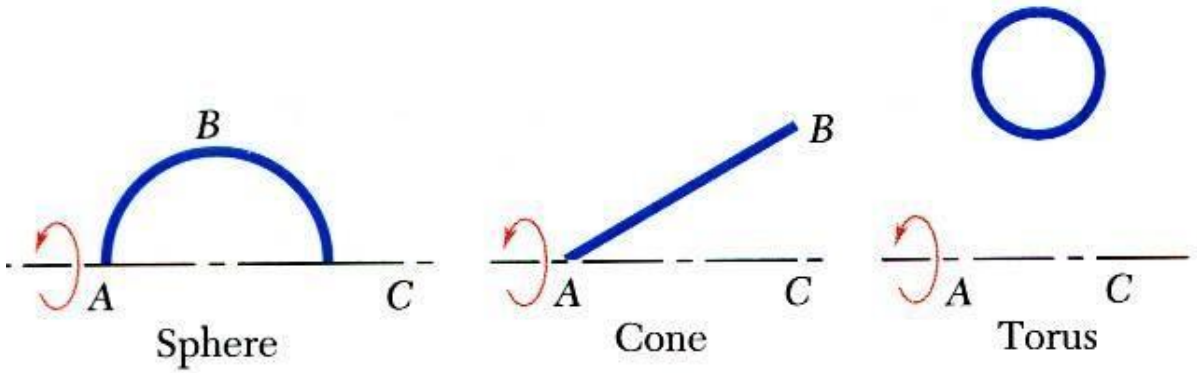
$$+\rightarrow \sum F_X = -A_x + 47.92 \sin 45^\circ = 0$$

$$A_X = 33.9\ kN$$

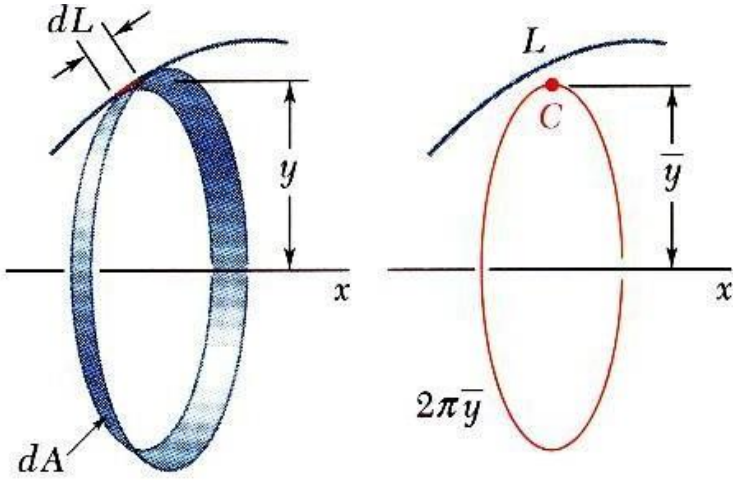
$$+\uparrow \sum F_Y = A_y + 47.92 \cos 45^\circ - 107.8 = 0$$

$$A_Y = 73.9\ kN$$

Pappus-Guldinus Teoremleri



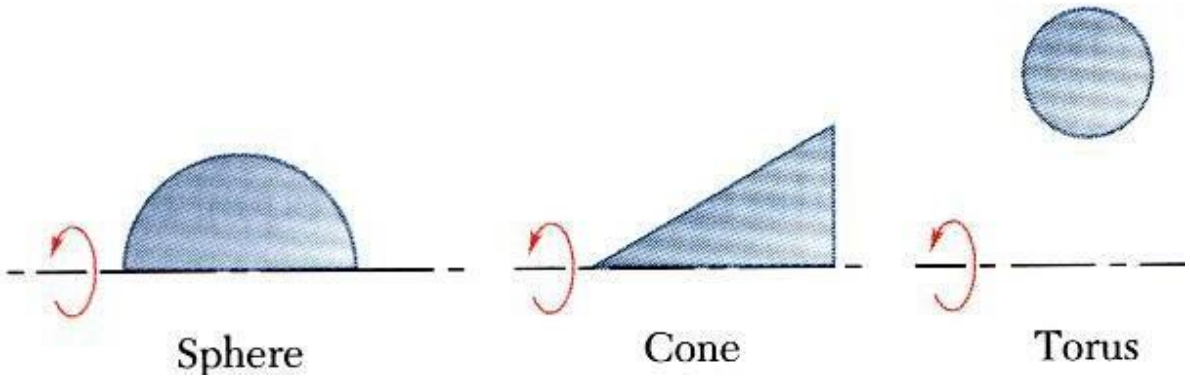
- Bir dönel yüzey, düzlemsel bir eğriyi sabit bir eksen etrafında döndürerek elde edilebilir.



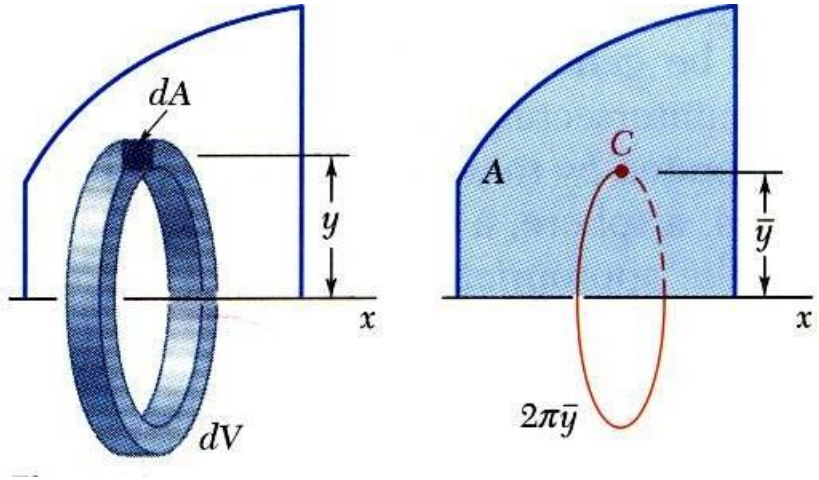
- Bir dönel yüzeyin alanı, kendisini meydana getiren eğrinin boyu ile yüzeyin meydana gelmesi sırasında eğrinin ağırlık merkezinin katettiği uzaklığın çarpımına eşittir.

$$A = 2\pi \bar{y} L$$

Pappus-Guldinus Teoremleri



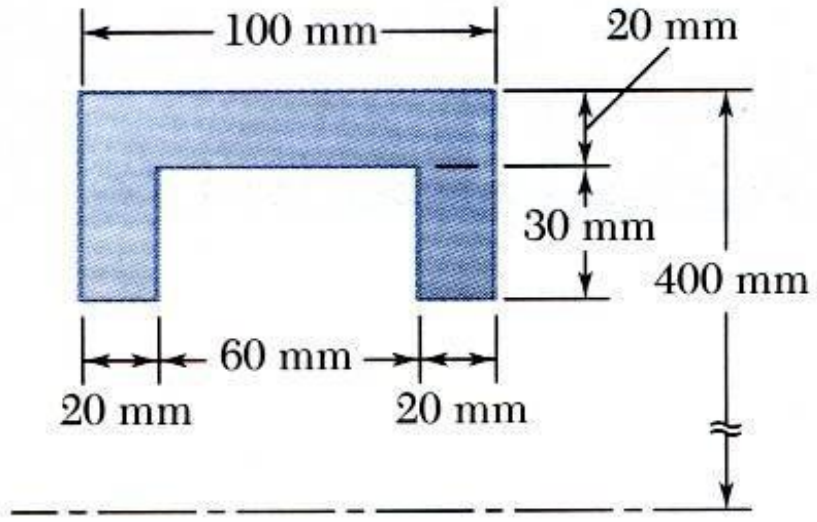
- Bir dönel cisim, düzlemsel bir alanın sabit bir eksen etrafında döndürerek elde edilebilir.



- Bir dönel cismin hacmi, kendisini meydana getiren alan ile cismin meydana gelmesi sırasında alanın ağırlık merkezinin katettiği uzaklığın çarpımına eşittir.

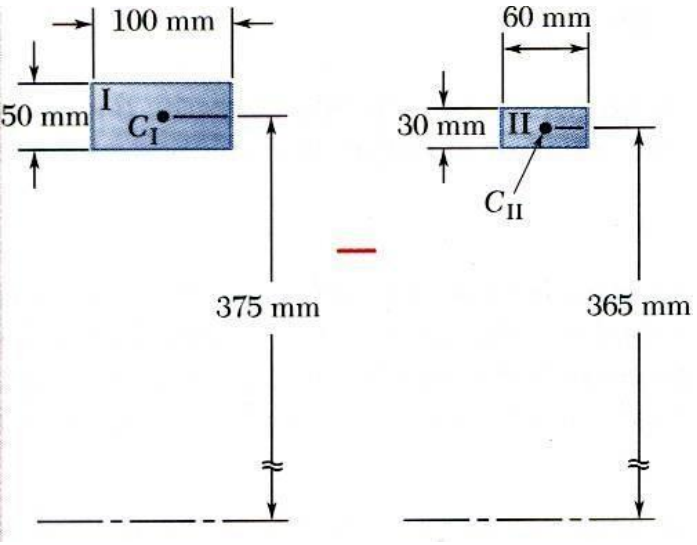
$$V = 2\pi \bar{y} A$$

Problem



Bir kasnağın dış çapı 0.8 m ve kesiti şekilde gösterildiği gibidir. Çelikten imal edilmiş kasnağın malzemesinin özgül ağırlığı bilindiğine göre kasnağın kütlesini ve ağırlığını hesaplayınız.

$$\rho = 7.85\times10^{3}\ \text{kg}/\text{m}^3$$



	Area, mm²	$\bar{y}$, mm	Distance Traveled by C, mm	Volume, mm³
I	+5000	375	$2\pi(375) = 2356$	$(5000)(2356) = 11.78\times10^{6}$
II	−1800	365	$2\pi(365) = 2293$	$(-1800)(2293) = -4.13\times10^{6}$
				Volume of rim $= 7.65\times10^{6}$

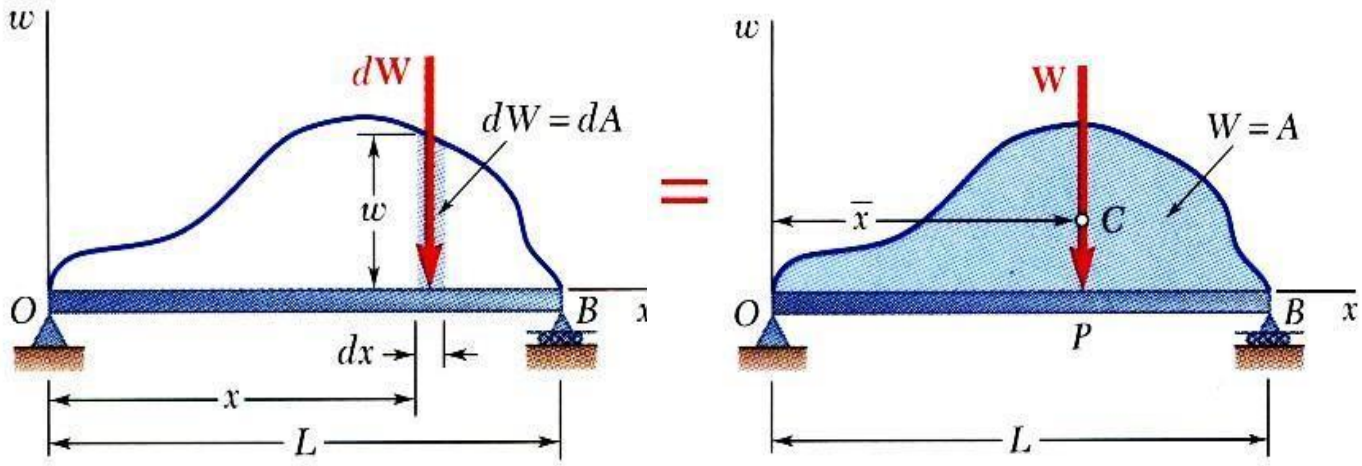
$$m = \rho V = \left(7.85\times10^{3}\ \text{kg}/\text{m}^3\right)\left(7.65\times10^{6}\ \text{mm}^3\right)\left(10^{-9}\ \text{m}^3/\text{mm}^3\right)$$

$$\boxed{m = 60.0\ \text{kg}}$$

$$W = mg = (60.0\ \text{kg})\left(9.81\ \text{m}/\text{s}^2\right)$$

$$\boxed{W = 589\ \text{N}}$$

Kirişlerde yayılı yükler



$$W = \int_0^L w\,dx = \int dA = A$$

- Yayılı yük birim uzunluğun taşıdığı yükü ifade eden, w (N/m) gösterilir. Toplam yük, yük eğrisi altında kalan alana eşittir.

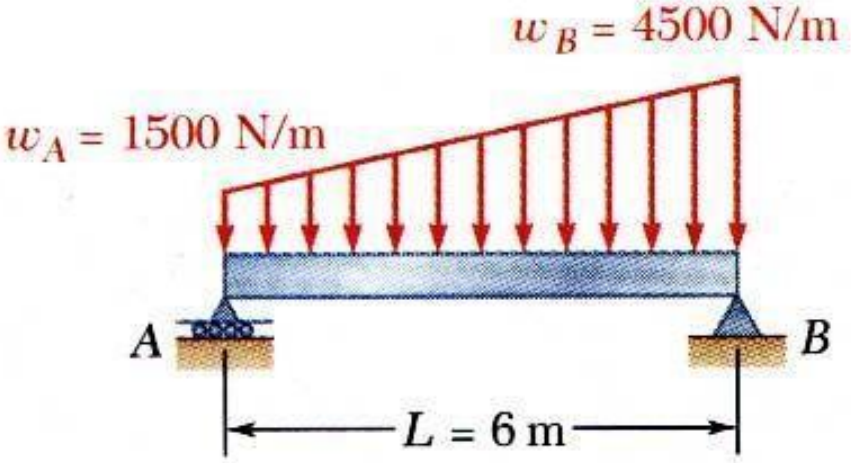
$$(OP)W = \int x\,dW$$

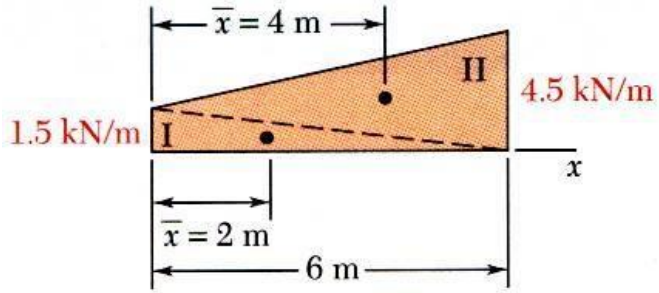
$$(OP)A = \int_0^L x\,dA = \bar{x}A$$

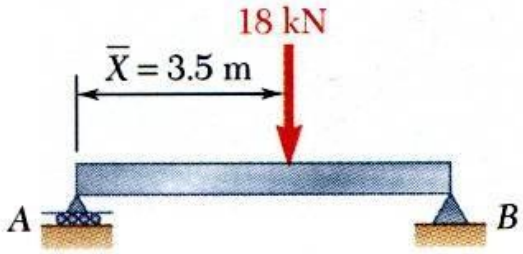
- Bir yayılı yük , şiddeti yük eğrisi altında kalan alana eşit, tesir çizgisi bu alanın ağırlık merkezinden geçen tekil bir yük olarak ele alınabilir.

Problem

Bir kiriş şekildeki gibi bir yayılı yük taşımaktadır. Eşdeğer münferit(tekil) yükü ve mesnet reaksiyonlarını bulunuz.







$$X = \frac{63\ \text{kN}\cdot\text{m}}{18\ \text{kN}}$$

$$\boxed{\bar{X} = 3.5\ \text{m}}$$

$$\sum M_A = 0: \quad B_y(6\,\text{m}) - (18\,\text{kN})(3.5\,\text{m}) = 0$$

$$\boxed{B_y = 10.5\,\text{kN}}$$

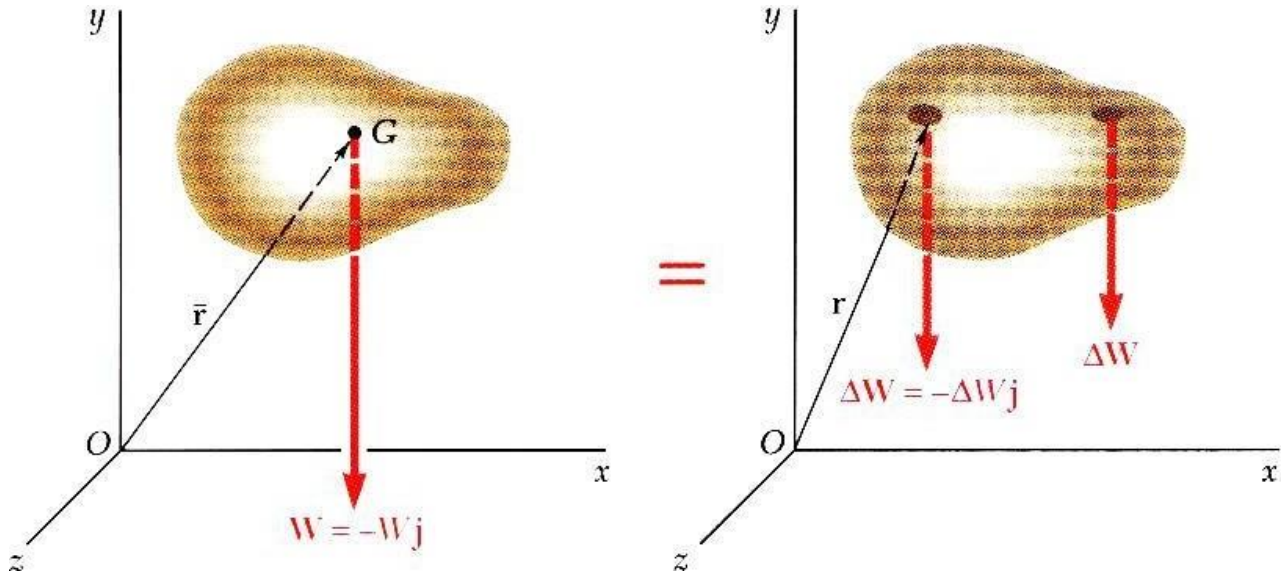
$$\boxed{F = 18.0\ \text{kN}}$$

$$\sum M_B = 0: \quad -A_y(6\,\text{m}) + (18\,\text{kN})(6\,\text{m} - 3.5\,\text{m}) = 0$$

$$\boxed{A_y = 7.5\ \text{kN}}$$

Component	A, kN	$\bar{x}$, m	$\bar{x}A$, kN·m
Triangle I	4.5	2	9
Triangle II	13.5	4	54
	$\Sigma A = 18.0$		$\Sigma \bar{x}A = 63$

3D bir cismin ağırlık merkezi: Bir hacmin merkezi



- Ağırlık merkezi yeri G

$$-W\vec{j} = \sum(-\Delta W\vec{j})$$

$$\vec{r}_G \times(-W\vec{j}) = \sum[\vec{r}\times(-\Delta W\vec{j})]$$

$$\vec{r}_G W \times(-\vec{j}) = (\sum \vec{r}\Delta W)\times(-\vec{j})$$

$$W = \int dW \qquad \vec{r}_G W = \int \vec{r}\, dW$$

- Cismin duruş şeklinden bağımsızdır,

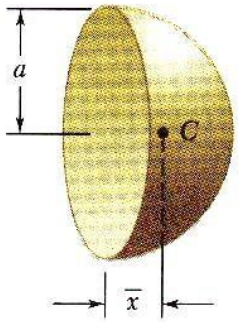
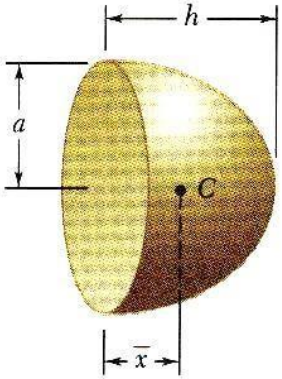
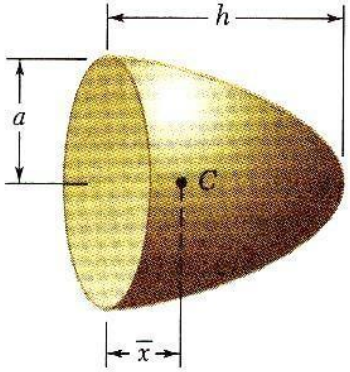
$$\bar{x}W = \int x\,dW \quad \bar{y}W = \int y\,dW \quad \bar{z}W = \int z\,dW$$

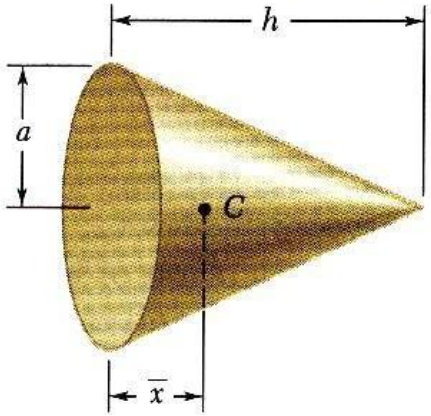
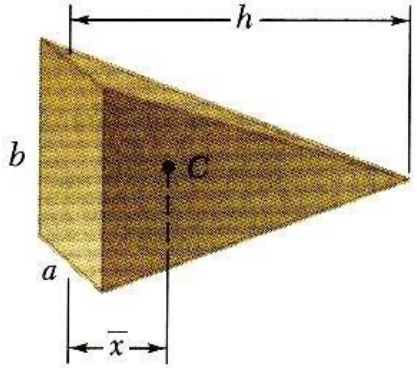
- Homojen bir cisim için,

$$W = \gamma V \text{ and } dW = \gamma\, dV$$

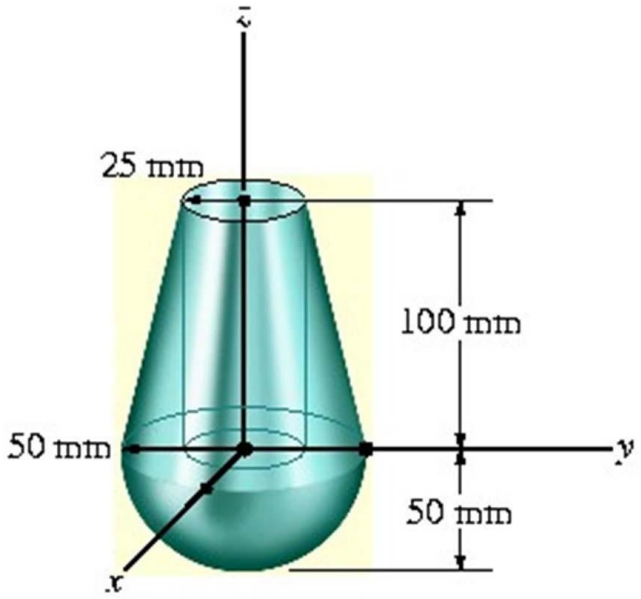
$$\bar{x}V = \int x\,dV \quad \bar{y}V = \int y\,dV \quad \bar{z}V = \int z\,dV$$

3D Hacim biçimlerinin ağırlık merkezi

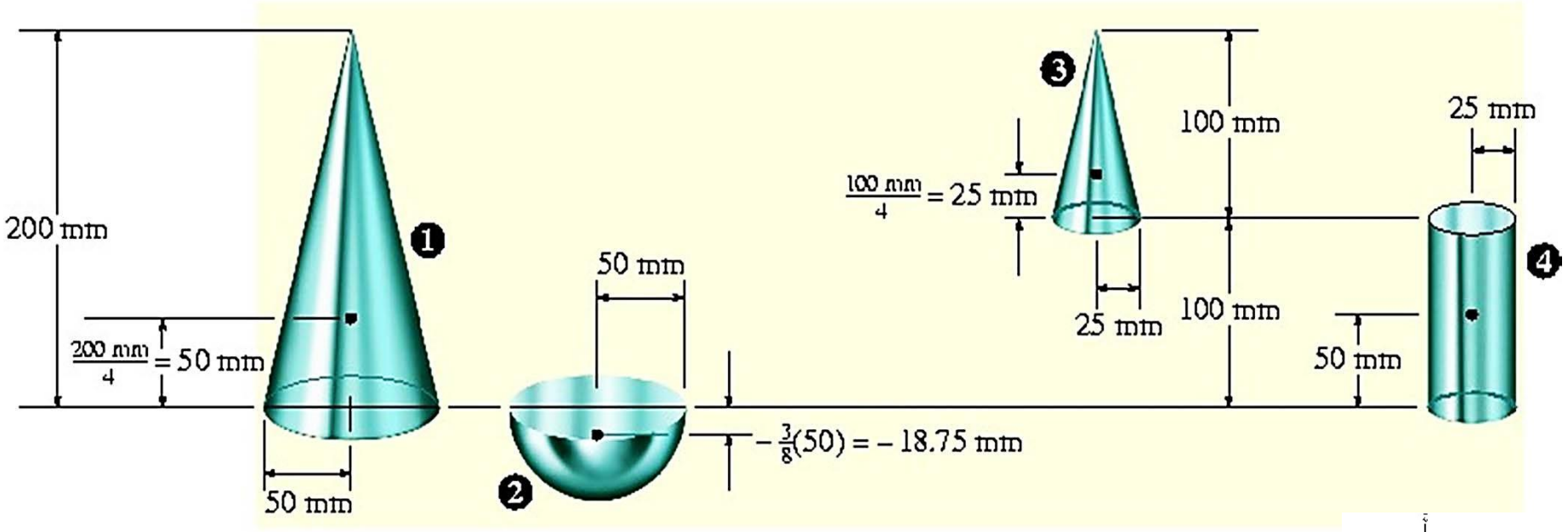
Shape		$\bar{x}$	Volume
Yarım küre		$\frac{3a}{8}$	$\frac{2}{3}\pi a^3$
Dönel Yarım elipsoit Semiellipsoid of revolution		$\frac{3h}{8}$	$\frac{2}{3}\pi a^2 h$
Dönel paraboloit Paraboloid of revolution		$\frac{h}{3}$	$\frac{1}{2}\pi a^2 h$

Koni		$\frac{h}{4}$	$\frac{1}{3}\pi a^2 h$
Pramit		$\frac{h}{4}$	$\frac{1}{3}abh$

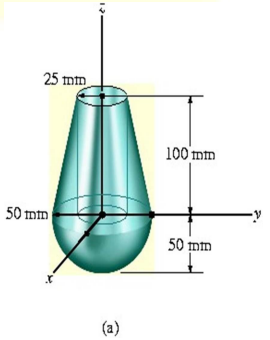
Şekildeki homojen dönel cismin ağırlık merkezini bulunuz. Kesik Konik'in özgül kütlesi $\rho = 8 Mg/m^3$ ve yarım kürenin özgül kütlesi $\rho = 4 Mg/m^3$ dir. Ayrıca merkezde 25 mm yarıçapında bir delik açılmıştır.



(a)



(b)



(a)

Parça	m (kg)	z (mm)	zm (kg.mm)
1	4.189	50	209.440
2	1.047	-18.75	-19.635
3	-0.524	125	-65.450
4	-1.571	50	-78.540
Toplam	3.141		45.815

$$\bar{z} = \frac{\sum \tilde{z} m}{\sum m} = \frac{45.815}{3.141} = 14.6mm$$